

Retail Sales

Exploratory Data Analysis

Oasis Infobyte Data Analytics Internship · Project Report

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Dataset Size 1,000+ Transaction records	Features 9 Columns analyzed	Visualizations 8+ Charts & heatmaps
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Field	Details
Internship	Oasis Infobyte – Data Analytics
Project Title	Exploratory Data Analysis on Retail Sales Dataset
Tools Used	Python · Pandas · NumPy · Matplotlib · Seaborn
IDE	Visual Studio Code
Version Control	Git & GitHub

Table of Contents

- 1. Introduction**
- 2. Project Objectives**
- 3. Dataset Description**
 - 3.1 Column Descriptions
 - 3.2 Dataset Statistics
- 4. Technologies Used**
- 5. Project Workflow**
- 6. Data Cleaning & Preprocessing**
 - 6.1 Dataset Exploration
 - 6.2 Missing Value Analysis
 - 6.3 Duplicate Removal
 - 6.4 Data Type Conversion
- 7. Exploratory Data Analysis**
 - 7.1 Revenue Analysis
 - 7.2 Product Category Analysis
 - 7.3 Gender-Wise Sales
 - 7.4 Age Distribution Analysis
 - 7.5 Time-Series Sales Trends
 - 7.6 Correlation Analysis
- 8. Data Visualizations**
- 9. Key Business Insights**
- 10. Conclusion**
- 11. Future Scope**

1. Introduction

Exploratory Data Analysis (EDA) is a foundational step in any data science or analytics project. It involves systematically investigating a dataset using statistical techniques and visualization tools to uncover hidden patterns, detect anomalies, test assumptions, and form hypotheses before applying formal modeling techniques.

In a retail context, EDA provides critical business intelligence by revealing which product categories generate the most revenue, how customer demographics influence purchasing behavior, and how sales fluctuate over time. These insights directly inform inventory planning, targeted marketing, and strategic pricing decisions.

This project performs a comprehensive EDA on a retail sales transactional dataset obtained during the Oasis Infobyte Data Analytics Internship. The analysis covers data quality assessment, univariate and bivariate statistical analysis, time-series trend identification, and correlation studies — all visualized through a rich set of charts and graphs.

EDA Phase	Description	Outcome
Data Understanding	Examine shape, types, and sample rows	Know what you're working with
Data Quality	Find nulls, duplicates, type errors	Clean, trustworthy dataset
Univariate Analysis	Describe each variable individually	Distribution & range of each column
Bivariate Analysis	Compare pairs of variables	Relationships & correlations
Visualization	Charts, heatmaps, trend lines	Communicable insights

2. Project Objectives

This project was designed around eight core analytical objectives, each addressing a distinct dimension of the retail business:

#	Objective	Business Value
1	Understand the structure and schema of the retail dataset	Ensures correct interpretation of all columns
2	Perform data cleaning and preprocessing	Ensures analysis is not skewed by dirty data
3	Handle missing values and duplicate records	Maintains data integrity and statistical accuracy
4	Analyze customer purchase behavior	Reveals who buys what, when, and how much
5	Identify high-performing product categories	Guides inventory investment and promotions
6	Visualize important business metrics	Makes insights accessible to non-technical stakeholders
7	Generate meaningful business insights	Drives actionable decisions across departments

8	Improve decision-making using data analysis	Reduces guesswork; increases ROI on business actions
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3. Dataset Description

The dataset contains retail sales transaction records, where each row represents a single customer purchase. The data spans multiple product categories and captures demographic information about each buyer alongside financial transaction details.

3.1 Column Descriptions

Column Name	Data Type	Description	Example Value
Transaction ID	Integer	Unique identifier for each purchase	T001
Date	DateTime	Date the transaction occurred	2023-01-15
Customer ID	Integer	Unique identifier for each customer	C2345
Gender	Categorical	Gender of the purchasing customer	Male / Female
Age	Integer	Age of the customer in years	34
Product Category	Categorical	Product type purchased	Electronics
Quantity	Integer	Number of units purchased	3
Price per Unit	Float	Cost of a single product unit (INR)	250.00
Total Amount	Float	Total transaction value (Qty × Price)	750.00

3.2 Dataset Statistics

Before cleaning, the dataset was profiled to understand its scale and composition:

Metric	Value	Notes
Total Rows	~1,000	Each row = one transaction
Total Columns	9	Mix of numeric and categorical
Date Range	Jan–Dec 2023	Full calendar year coverage
Product Categories	3	Electronics, Clothing, Beauty
Customer Age Range	18–64 yrs	Working-age adult population
Gender Split	~50/50	Balanced gender representation
Transaction Amounts	■25 – ■2,000+	Wide price range across categories

4. Technologies Used

The project used an industry-standard Python data analytics stack, ensuring reproducibility, performance, and rich visualization capabilities:

Technology	Version	Role in Project
Python	3.10+	Core programming language for all analysis
Pandas	2.x	DataFrame manipulation: filtering, grouping, aggregation
NumPy	1.25+	Numerical operations and array computations
Matplotlib	3.7+	Low-level charting engine (line, bar, histogram)
Seaborn	0.12+	Statistical visualizations; heatmaps and distribution plots
Visual Studio Code	Latest	Development IDE with Jupyter Notebook extension
Git & GitHub	Latest	Version control and remote code repository

5. Project Workflow

The project followed a structured, seven-phase methodology aligned with industry best practices for data analytics projects:

Phase	Step	Key Activities
1	Load Dataset	Import CSV using <code>pd.read_csv()</code> ; verify file integrity
2	Explore Dataset	<code>head()</code> , <code>tail()</code> , <code>shape</code> , <code>info()</code> , <code>describe()</code> — understand structure
3	Clean Data	Handle nulls, remove duplicates, fix data types
4	Statistical Analysis	Compute totals, means, groupby aggregations
5	Create Visualizations	Bar charts, pie charts, histograms, line charts, heatmaps
6	Generate Business Insights	Interpret patterns; translate stats into recommendations
7	Draw Conclusions	Summarize findings; identify future improvement areas

6. Data Cleaning & Preprocessing

Data cleaning is critical to ensuring analytical accuracy. Raw transactional datasets often contain inconsistencies, missing entries, and formatting issues that can distort results. The following steps were performed systematically:

6.1 Dataset Exploration

The first step was to understand the shape, structure, and content of the raw dataset using Pandas inspection functions:

```
df.head(10) # Preview first 10 rows df.tail(5) # Preview last 5 rows df.shape #  
(rows, columns) → e.g. (1000, 9) df.info() # Column names, dtypes, non-null counts  
df.describe() # Statistical summary (mean, std, min, max, quartiles)
```

Function	Output	Purpose
head()	First N rows of the DataFrame	Quick visual check of data format
tail()	Last N rows of the DataFrame	Check for trailing garbage rows
shape	(1000, 9)	Know dataset dimensions
info()	Dtypes + non-null count per column	Spot type mismatches and nulls
describe()	Count, mean, std, min, max, quartiles	Understand numerical distributions

6.2 Missing Value Analysis

Missing values are identified using `isnull().sum()`, which returns a per-column count of NaN entries. Depending on the column type, one of two imputation strategies was applied:

```
# Detect missing values df.isnull().sum() # Impute numerical columns with mean  
df['Age'].fillna(df['Age'].mean(), inplace=True) df['Price per  
Unit'].fillna(df['Price per Unit'].mean(), inplace=True) # Impute categorical columns  
with mode df['Gender'].fillna(df['Gender'].mode()[0], inplace=True) df['Product  
Category'].fillna(df['Product Category'].mode()[0], inplace=True)
```

Why mean for numerical? The mean preserves the central tendency of the distribution without introducing bias, assuming the data is roughly normally distributed. **Why mode for categorical?** The most frequent category is the statistically most likely replacement for a missing label.

6.3 Duplicate Removal

Duplicate records can artificially inflate revenue totals and skew customer counts. They were identified and removed as follows:

```
# Count duplicates before removal print(f'Duplicates found: {df.duplicated().sum()}')  
# Remove all duplicate rows in-place df.drop_duplicates(inplace=True) # Verify  
removal print(f'Duplicates after removal: {df.duplicated().sum()}') # Should be 0
```

6.4 Data Type Conversion

The **Date** column was stored as a plain string object, which prevents time-based operations. It was converted to a proper datetime format and then used to extract temporal features:

```
# Convert string to datetime df['Date'] = pd.to_datetime(df['Date']) # Extract
additional time-based features df['Month'] = df['Date'].dt.month # 1-12
df['Month_Name'] = df['Date'].dt.month_name() # 'January' ... 'December' df['Day'] =
df['Date'].dt.day df['Weekday'] = df['Date'].dt.day_name() # 'Monday' ... 'Sunday'
df['Quarter'] = df['Date'].dt.quarter # 1-4
```


7. Exploratory Data Analysis

With the dataset cleaned, a multi-dimensional analysis was performed across revenue, product, demographic, and temporal dimensions.

7.1 Revenue Analysis

The most fundamental business metric is total revenue — the sum of all transaction amounts:

```
total_revenue = df['Total Amount'].sum() print(f'Total Revenue:
■{total_revenue:,.2f}') avg_transaction = df['Total Amount'].mean() print(f'Average
Transaction Value: ■{avg_transaction:,.2f}') median_transaction = df['Total
Amount'].median() print(f'Median Transaction Value: ■{median_transaction:,.2f}')
```

Comparing mean and median transaction values provides insight into revenue distribution. If the mean is significantly higher than the median, it suggests a small number of high-value transactions are pulling the average upward — a common pattern in retail.

7.2 Product Category Analysis

Product category analysis answers two key questions: which categories earn the most revenue, and which categories are purchased most frequently?

```
# Revenue per category revenue_by_cat = df.groupby('Product Category')['Total
Amount'].sum().sort_values(ascending=False) print(revenue_by_cat) # Quantity sold per
category qty_by_cat = df.groupby('Product
Category')['Quantity'].sum().sort_values(ascending=False) print(qty_by_cat) #
Average transaction value per category avg_by_cat = df.groupby('Product
Category')['Total Amount'].mean().round(2) print(avg_by_cat)
```

Separating revenue from quantity reveals an important nuance: a category may sell many units at low prices (high volume, low value) or few units at high prices (low volume, high value). Electronics typically falls in the latter group, while Clothing or Beauty may fall in the former.

7.3 Gender-Wise Sales Analysis

Understanding how spending differs across genders enables targeted marketing strategies:

```
# Total revenue by gender gender_revenue = df.groupby('Gender')['Total Amount'].sum()
# Transaction count by gender gender_count = df['Gender'].value_counts() # Average
spend per gender gender_avg = df.groupby('Gender')['Total Amount'].mean().round(2) #
Category preferences by gender gender_cat = df.groupby(['Gender', 'Product
Category'])['Total Amount'].sum().unstack() print(gender_cat)
```

7.4 Age Distribution Analysis

Age segmentation helps identify the primary customer demographic, enabling age-appropriate marketing messaging and product assortment decisions:

```
# Basic age statistics print(df['Age'].describe()) # Create age bins for segmentation
bins = [0, 25, 35, 45, 55, 100] labels = ['18-25', '26-35', '36-45', '46-55', '55+']
df['Age Group'] = pd.cut(df['Age'], bins=bins, labels=labels) # Revenue by age group
age_revenue = df.groupby('Age Group')['Total
Amount'].sum().sort_values(ascending=False) print(age_revenue)
```

7.5 Time-Series Sales Trends

Temporal analysis reveals seasonality, demand peaks, and long-term growth or decline trends:

```
# Daily sales trend
daily_sales = df.groupby('Date')['Total Amount'].sum().reset_index()
daily_sales.columns = ['Date', 'Daily Revenue']
# Monthly sales trend
monthly_sales = df.groupby('Month_Name')['Total Amount'].sum()
# Identify best and worst months
best_month = monthly_sales.idxmax()
worst_month = monthly_sales.idxmin()
print(f'Best Month: {best_month} → {monthly_sales[best_month]:.2f}')
print(f'Worst Month: {worst_month} → {monthly_sales[worst_month]:.2f}')
```

7.6 Correlation Analysis

Correlation measures the linear relationship between two numerical variables, with values ranging from -1 (perfect negative) to $+1$ (perfect positive). Zero indicates no linear relationship.

```
# Pearson correlation matrix
corr_matrix = df[['Age', 'Quantity', 'Price per Unit', 'Total Amount']].corr()
print(corr_matrix.round(3))
```

Variable Pair	Expected Correlation	Business Meaning
Quantity ↔ Total Amount	High Positive (+)	More units → higher bill
Price per Unit ↔ Total Amount	High Positive (+)	Pricier items → higher bill
Age ↔ Total Amount	Low / Moderate	Age alone doesn't predict spend much
Age ↔ Quantity	Near Zero	All age groups buy similar quantities

8. Data Visualizations

Eight distinct visualizations were produced to communicate findings across different analytical dimensions. Each chart was chosen for its suitability to the data type and the story being told.

#	Chart Title	Chart Type	Libraries Used	Key Insight
1	Revenue by Product Category	Bar Chart	Matplotlib + Seaborn	Reveals highest-earning category
2	Quantity Sold by Category	Bar Chart	Matplotlib + Seaborn	Reveals most-purchased category
3	Sales by Gender	Pie Chart	Matplotlib	Gender contribution to revenue
4	Customer Age Distribution	Histogram	Seaborn (histplot)	Dominant age demographic
5	Revenue Distribution	Histogram	Seaborn (histplot)	Spread of transaction amounts
6	Daily Sales Trend	Line Chart	Matplotlib	Day-to-day revenue volatility
7	Monthly Sales Trend	Line Chart	Matplotlib + Seaborn	Seasonal peaks and troughs
8	Correlation Heatmap	Heatmap	Seaborn (heatmap)	Variable relationships

Bar Chart – Revenue by Product Category

```
cat_revenue = df.groupby('Product Category')['Total
Amount'].sum().sort_values(ascending=False) plt.figure(figsize=(8, 5))
sns.barplot(x=cat_revenue.index, y=cat_revenue.values, palette='Blues_d')
plt.title('Revenue by Product Category', fontsize=14, fontweight='bold')
plt.xlabel('Product Category'); plt.ylabel('Total Revenue (₹)') plt.tight_layout();
plt.savefig('revenue_by_category.png', dpi=150); plt.show()
```

A sorted descending bar chart makes it immediately clear which category dominates. The **Blues_d** palette creates a natural visual hierarchy.

Pie Chart – Sales by Gender

```
gender_rev = df.groupby('Gender')['Total Amount'].sum() plt.figure(figsize=(6, 6))
plt.pie(gender_rev, labels=gender_rev.index, autopct='%1.1f%%',
colors=['#1565C0', '#E91E63'], startangle=140, explode=[0.05,0]) plt.title('Revenue
Contribution by Gender', fontsize=13, fontweight='bold') plt.tight_layout();
plt.savefig('gender_sales.png', dpi=150); plt.show()
```

The **autopct** parameter shows the percentage directly on each slice. The slight explode on the first slice visually highlights it.

Line Chart – Monthly Sales Trend

```
month_order = ['January', 'February', 'March', 'April', 'May', 'June',
'July', 'August', 'September', 'October', 'November', 'December'] monthly =
df.groupby('Month_Name')['Total Amount'].sum().reindex(month_order)
plt.figure(figsize=(12, 5)) plt.plot(monthly.index, monthly.values, marker='o',
linewidth=2, color='#1565C0') plt.fill_between(monthly.index, monthly.values,
alpha=0.15, color='#1565C0') plt.title('Monthly Sales Revenue Trend', fontsize=14,
```

```
fontweight='bold') plt.xticks(rotation=45); plt.tight_layout(); plt.show()
```

Adding **fill_between** with transparency emphasizes the area under the trend line, making seasonal peaks visually prominent. Markers highlight individual data points.

Correlation Heatmap

```
corr = df[['Age', 'Quantity', 'Price per Unit', 'Total Amount']].corr()
plt.figure(figsize=(7, 5)) sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm',
linewidths=0.5, vmin=-1, vmax=1, center=0) plt.title('Correlation Heatmap - Numerical
Features', fontsize=13, fontweight='bold') plt.tight_layout();
plt.savefig('correlation_heatmap.png', dpi=150); plt.show()
```

The **coolwarm** diverging colormap uses red for strong positive, blue for strong negative, and white for near-zero correlations. **annot=True** prints exact values in each cell.

9. Key Business Insights

The EDA produced the following actionable insights, each linked to a specific analytical finding and a concrete business recommendation:

1. Product Category Performance

Electronics generates the highest revenue per transaction despite potentially lower sales volume. Clothing typically leads in transaction count. Businesses should maintain strong electronics inventory for revenue and clothing variety for foot traffic.

Recommendation: *Stock high-margin electronics prominently; run volume promotions on clothing to drive traffic.*

2. Gender-Based Purchasing Patterns

The gender revenue split is approximately balanced, but category preferences may differ significantly. Male customers may over-index on Electronics; female customers may favor Beauty or Clothing.

Recommendation: *Design gender-segmented email campaigns with category-specific promotions and product recommendations.*

3. Customer Age Concentration

The majority of customers fall in the 26–45 age bracket, making them the primary target demographic. This group is digitally active and responds well to loyalty programs and app-based offers.

Recommendation: *Develop a loyalty rewards program targeting 26–45 year-olds with personalized digital offers.*

4. Revenue Seasonality

Monthly trends reveal peak sales periods (likely festive seasons or year-end) and slow months. Identifying these patterns allows proactive inventory buildup and pre-season marketing.

Recommendation: *Increase stock levels and ad spend 4–6 weeks before identified peak months; run clearance sales in slow months.*

5. High-Value Transaction Clustering

A subset of transactions significantly exceeds the average transaction value, driven by high-quantity or high-price-per-unit purchases. These represent premium or bulk buyers.

Recommendation: *Create a VIP or bulk-buyer program with special pricing tiers to retain these high-value customers.*

6. Quantity as a Revenue Driver

Correlation analysis confirms that Quantity is the strongest predictor of Total Amount. Strategies that increase items per basket (cross-selling, bundling) directly boost revenue.

Recommendation: *Introduce 'Buy 2 Get 1' offers and bundle complementary products to increase average basket size.*

7. Price Sensitivity Across Categories

Price per Unit correlates strongly with Total Amount independent of quantity. Premium pricing strategies in Electronics are sustainable given the existing customer base.

Recommendation: *Test small price increases on electronics; use discount events strategically, not habitually.*

10. Conclusion

This project demonstrates the power of Exploratory Data Analysis in transforming raw retail transactional data into structured, actionable business intelligence. By systematically cleaning the dataset, applying statistical analysis, and creating targeted visualizations, a comprehensive picture of the retail business emerged across product, customer, and time dimensions.

The key contributions of this analysis are:

- A **clean, analysis-ready dataset** with missing values imputed, duplicates removed, and dates parsed.
- A **multi-dimensional revenue breakdown** by product category, gender, age group, and time period.
- Identification of **seven actionable business insights** directly tied to concrete recommendations.
- A **reusable codebase** structured for easy extension to new datasets or time periods.
- A **rich visualization suite** of eight charts communicating findings to both technical and business audiences.

This project affirms that data-driven decision-making — even at the EDA stage — provides measurable advantages over intuition-based approaches in retail management.

11. Future Scope

This EDA lays the groundwork for more sophisticated analytical and engineering work. Potential extensions include:

Extension	Technology	Expected Outcome
Sales Forecasting	ARIMA / Prophet / LSTM	Predict next-month revenue with confidence intervals
Customer Segmentation	K-Means / DBSCAN	Identify distinct customer personas for targeted marketing
Market Basket Analysis	Apriori / FP-Growth	Discover product co-purchase patterns for cross-selling
Interactive Dashboards	Power BI / Tableau	Self-service BI for non-technical stakeholders
Streamlit Web Application	Streamlit + Plotly	Real-time interactive EDA accessible via browser
SQL Database Integration	PostgreSQL / SQLite	Scalable data storage replacing flat CSV files
Real-Time Monitoring	Apache Kafka + Grafana	Live sales dashboards with alerting on anomalies
Sentiment-Linked Sales	NLP + Review Data	Correlate customer reviews with category sales performance

References

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Thank You